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Ext 1

NORTH SYDNEY BOYS HIGH SCHOOL

2005
TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write on one side of the paper (with lines) in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a new page.

- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Lowe
- ☐ Mr Rezcallah
- ☐ Mr Trenwith
- ☐ Mr Ee
- ☐ Ms Silverman
- ☐ Mr Weiss

Student Number: _____

(To be used by the exam markers only.)

Question No	1	2	3	4	5	6	7	Total	Total
Mark	$\frac{\quad}{12}$	$\frac{\quad}{12}$	$\frac{\quad}{12}$	$\frac{\quad}{12}$	$\frac{\quad}{12}$	$\frac{\quad}{12}$	$\frac{\quad}{12}$	$\frac{\quad}{84}$	$\frac{\quad}{100}$

Question 1

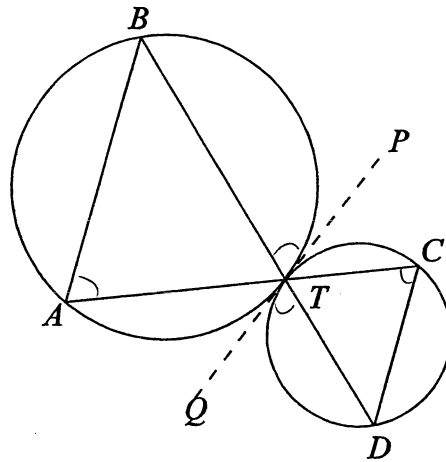
- (a) Differentiate $e^x \cos 2x$. 2
- (b) Find the acute angle between the lines $y = 2x + 3$ and $3x - 2y - 1 = 0$.
Write your answer correct to the nearest degree. 2
- (c) Use the table of standard integrals to evaluate $\int_3^5 \frac{dx}{\sqrt{x^2 - 9}}$. 3
Write your answer in the form $\ln a$, where a is a constant.
- (d) A and B are the points $(5, 1)$ and $(-1, 4)$ respectively. 2
Find the coordinates of the point P which divides AB externally in the ratio $5 : 2$.
- (e) Evaluate $\int_{\frac{1}{3}}^{\frac{2}{3}} 9x(3x - 1)^4 dx$ using the substitution $u = 3x - 1$. 3

Question 2 (Start a new page)

- (a) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$ 1
- (b) α, β, γ are the roots of the polynomial equation $2x^3 - 3x + 2 = 0$. Evaluate
- (i) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ 2
- (ii) $\alpha^2 + \beta^2 + \gamma^2$ 2
- (c) Assuming $x = 2$ is a close approximation to a root of $2 \sin x = x$, use one application of Newton's method to find a better approximation. Give your answer correct to three decimal places. 2
- (d) Find
- (i) $\int \sin^2 2x dx$ 2
- (ii) $\int_0^{\frac{4}{3}} \frac{2 dx}{16 + 9x^2}$ 3

Question 3 (Start a new page)

(a)



Two circles touch externally at T . PQ is a tangent to each circle at T . AB and CD are chords in the respective circles. ATC and BTD are straight lines.

(i) State why $\angle QTD = \angle TCD$.

1

(ii) Prove that $AB \parallel CD$.

3

(b) (i) Write $\cos x - \sqrt{3} \sin x$ in the form $R \cos(x + \alpha)$, where $0 \leq \alpha \leq \frac{\pi}{2}$.

2

(ii) Hence, or otherwise, solve $\cos x - \sqrt{3} \sin x = \sqrt{3}$ for $0 \leq x \leq 2\pi$.

2

(c) (i) Write a simplified expression for $2 \sin 2A \cos 2A$.

1

(ii) Prove the identity $\frac{\sin 3A}{\sin A} + \frac{\cos 3A}{\cos A} = 4 \cos 2A$.

3

[The result in part (b) (i) may be useful]

Question 4 (Start a new page)

- (a) Sketch the graph of $y = 2 \sin^{-1}(x - 1)$.

2

- (b) A piece of cork moves vertically in Simple Harmonic Motion on the surface of the water as waves pass under it. Its velocity v m/s is given by $v^2 = -x^2 + 7x - 12$, where x is the cork's vertical displacement in metres.

- (i) Using $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$, find the acceleration of the cork in terms of x .

1

- (ii) What is the centre of motion?

1

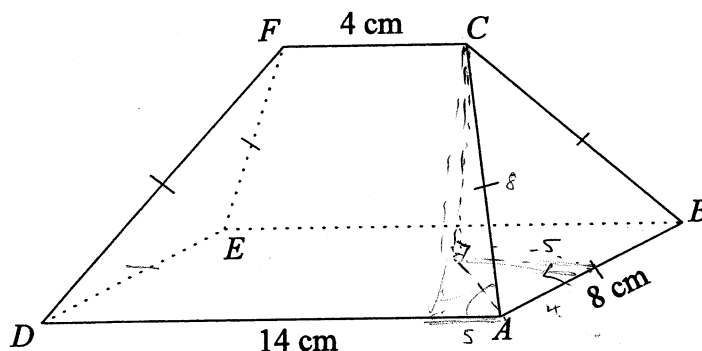
- (iii) Find the period of oscillation.

1

- (c) Solve for x : $\frac{2}{x-1} < x$

3

- (d)



In the solid shown, $ABED$ is a rectangle of length 14 cm and breadth 8 cm. ABC and DEF are congruent equilateral triangles of side length 8 cm. $ACFD$ and $BCFE$ are congruent isosceles trapezia, whose parallel sides are 14 cm and 4 cm, as shown. Find

- (i) the angle in the trapezium between AC and AD .

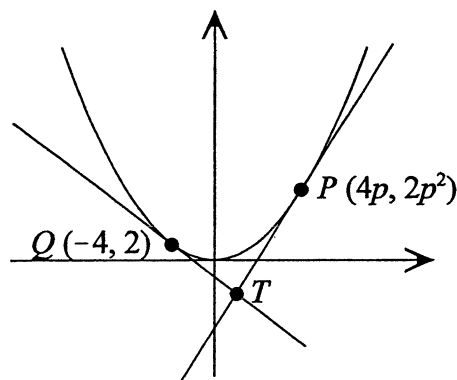
2

- (ii) the angle between AC and the base $ABED$.

2

Question 5 (Start a new page)

(a)



$P(4p, 2p^2)$ and $Q(-4, 2)$ are two points on the parabola $x^2 = 8y$.
The tangents to the parabola at P and Q intersect at T .

- (i) Show that the tangent at P has equation $y = px - 2p^2$. 2
- (ii) Hence, write down the equation of the tangent at Q . 1
- (iii) Show that T has coordinates $(2p - 2, -2p)$. 3
- (iv) M is the midpoint of P and T . 3

Show that the locus of M , as P varies on the parabola, has equation

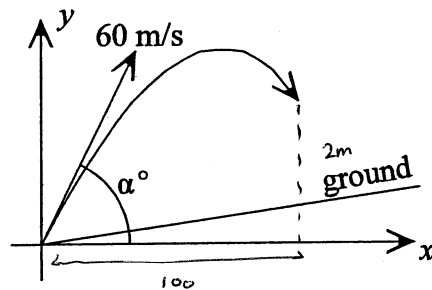
$$9y = (x + 1)(x - 2)$$

- (b) A cup of soup with a temperature of 95°C is placed in a room which has a temperature of 20°C . After 10 minutes, the soup cools to 70°C . The rate of heat loss is proportional to the difference between the soup's temperature and room temperature, that is $\frac{dT}{dt} = -k(T - 20)$.

- (i) Show that $T = 20 + Ae^{-kt}$ is a solution of this differential equation 1
- (ii) Find the temperature of the soup after a further 5 minutes, correct to the nearest degree. 2

Question 6 (Start a new page)

(a)



At the Battle of Hastings, the Normans fired arrows at the Anglo-Saxons up a hill which had a gradient of 1 in 10. The diagram shows the path of the arrows (assume that the arrows were fired from ground level). All arrows were fired with an initial velocity of 60 m/s. The archers varied the range by varying the angle of projection, α . Assume that the acceleration due to gravity is 10 m/s^2 .

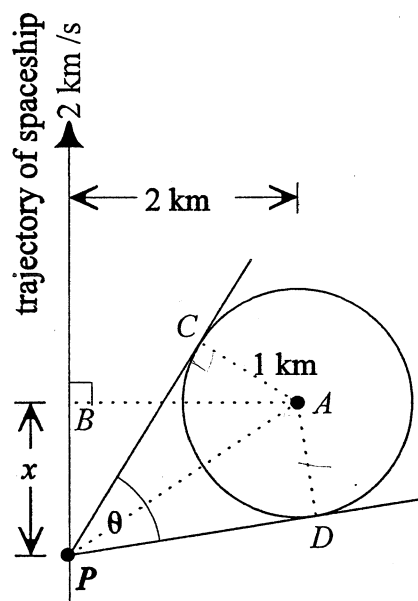
- (i) Show that the equations for horizontal and vertical displacement of an arrow are respectively $x = 60t \cos \alpha$ and $y = -5t^2 + 60t \sin \alpha$, where t is the time in seconds after firing the arrow. 4
 - (ii) Show that the Cartesian equation for the path of an arrow is 2

$$y = -\frac{1}{720}x^2(1 + \tan^2 \alpha) + x \tan \alpha.$$
 - (iii) According to legend Harold, King of the Anglo-Saxons, was killed when hit in the eye by a Norman arrow. Assume that Harold was 100 metres horizontally from the Norman archers, and that his eye was 2 metres above the ground. At what angle(s), α , must this arrow have been fired if it hit Harold on the way down. 3
- (b) When the polynomial $P(x)$ is divided by $x - 4$, the remainder is -5 . 3
 When $P(x)$ is divided by $x + 1$, the remainder is 5 .
 Find the remainder when $P(x)$ is divided by $(x - 4)(x + 1)$.

Question 7 (Start a new page)

- (a) (i) By using the formula for the sum of an arithmetic series, show that 1
- $$1 + 2 + 3 + \dots + n + (n+1) = \frac{(n+1)(n+2)}{2}$$
- (ii) Use mathematical induction to prove 4
- $$\frac{1}{1} + \frac{1}{1+2} + \frac{1}{1+2+3} + \dots + \frac{1}{1+2+3+\dots+n} = \frac{2n}{n+1} \text{ for } n \geq 1.$$
- The result of part (i) may be useful.
- (iii) Hence, write down the limiting sum of a series whose general term is given 1
- by $T_n = \frac{1}{1+2+3+\dots+n}$.

(b)



The diagram shows a spaceship flying past an asteroid. The asteroid has a radius of 1 km, and the spaceship is 2 km from the asteroid's centre at its closest approach.

When the spaceship is at the point P , it is x km from its closest approach. At this moment, the asteroid subtends an angle of θ radians at the spaceship.

The spaceship is travelling in a straight line at a constant speed of 2 km/s.

- (i) Show that the angle θ is given by $\theta = 2 \sin^{-1} \frac{1}{\sqrt{4+x^2}}$. 2
- (ii) At what rate, in degrees per second, is the angle θ changing when x is 3 km? 4

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

b) (i) $AP^2 = AB^2 + BA^2$

$AP = \sqrt{4+x^2}$ [1]

$\sin APC = \frac{1}{\sqrt{4+x^2}}$

$\hat{APC} = \sin^{-1} \frac{1}{\sqrt{4+x^2}}$

$\theta = 2 \sin^{-1} \frac{1}{\sqrt{4+x^2}}$ [1] [2]

(ii) $\theta = 2 \sin^{-1} (4+x^2)^{-1/2}$

$\frac{d\theta}{dx} = \frac{2}{\sqrt{1 - \frac{1}{4+x^2}}} \cdot \left(-\frac{1}{2}\right) (4+x^2)^{-3/2} \cdot 2x$ [1] for correct derivative (unsimplified)

$= \frac{-2x(4+x^2)^{-3/2}}{\sqrt{\frac{4+x^2-1}{4+x^2}}}$

$= \frac{-2x(4+x^2)^{-3/2} \cdot (4+x^2)^{1/2}}{(3+x^2)^{1/2}}$

$= -\frac{2x}{(4+x^2)\sqrt{3+x^2}}$

$\frac{d\theta}{dt} = \frac{d\theta}{dx} \cdot \frac{dx}{dt}$
 $= \frac{-4x}{(4+x^2)\sqrt{3+x^2}}$ [1] for $\frac{d\theta}{dt}$

$x=3, \frac{d\theta}{dt} = \frac{-12}{13\sqrt{12}}$ rad/s [1] for correct substitution
 $= 15.3^\circ/\text{sec}$ [1] for conversion

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Question 1

(a) $\frac{d}{dx} (e^x \cos 2x) = \cos 2x \cdot e^x + e^x \cdot (-2 \sin 2x)$ [2]
 $= e^x \cos 2x - 2e^x \sin 2x$
 $= e^x (\cos 2x - 2 \sin 2x)$

[1] for correctly using product rule

[1] for correctly differentiating e^x and $\cos 2x$ (but only if product rule correct)

(b) $y = 2x + 3 \Rightarrow m_1 = 2$

$3x - 2y - 1 = 0 \Rightarrow y = \frac{3}{2}x - \frac{1}{2} \Rightarrow m_2 = \frac{3}{2}$

$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$

$= \left| \frac{\frac{3}{2} - 2}{1 + \frac{3}{2} \cdot 2} \right|$ [1] for correct substitution into correct formula

$= \frac{1}{8}$

$\theta = \tan^{-1} \frac{1}{8}$

[1] for correct answer (ignore rounding) [2]

(c) $\int_3^5 \frac{dx}{\sqrt{x^2 - 9}} = \left[\ln \left(x + \sqrt{x^2 - 9} \right) \right]_3^5$ [1]

$= \ln(5+4) - \ln(3+0)$ [1]

$= \ln 9 - \ln 3$

$= \ln 3$

[1] [3]

$$m:n = 5:-2$$

$$x = \frac{nx_1 + mx_2}{m+n}$$

$$= \frac{(-2)(5) + 5(-1)}{5-2}$$

$$= -5$$

$$P(-5, 6)$$

$$y = \frac{ny_1 + my_2}{m+n}$$

$$= \frac{(-2)(1) + 5(4)}{5-2}$$

$$= 6$$

$$2) \int_{1/3}^{2/3} 9x(3x-1)^4 dx$$

$$u = 3x - 1$$

$$du = 3 dx$$

$$\text{At } x = \frac{1}{3}, u = 0$$

$$x = \frac{2}{3}, u = 1$$

$$= \int_{1/3}^{2/3} 3x \cdot (3x-1)^4 \cdot 3 dx$$

$$= \int_0^1 (u+1) \cdot u^4 \cdot du$$

$$[2] - [1] \text{ for correct integral}$$

$$[2] - [1] \text{ for correct limits}$$

$$= \int_0^1 (u^5 + u^4) du$$

$$= \left[\frac{u^6}{6} + \frac{u^5}{5} \right]_0^1$$

$$= \left(\frac{1}{6} + \frac{1}{5} \right) - 0$$

$$= \frac{11}{30}$$

$$[1] \text{ for final answer}$$

$$[3]$$

Marks 1 for making x & u in same

integral, or for not changing limits in the first

line where the variable was changed

Question 7

$$(a) \quad (i) \quad S_n = \frac{n}{2}(a+d)$$

$$\therefore 1+2+3+\dots+n+n+1 = \frac{n+1}{2}[1+(n+1)]$$

$$= \frac{(n+1)(n+2)}{2} \quad [1]$$

$$(ii) \quad \text{Prove: } \frac{1}{1} + \frac{1}{1+2} + \frac{1}{1+2+3} + \dots + \frac{1}{1+2+3+\dots+n} = \frac{2n}{n+1}$$

$$\text{LHS } n=1: \text{ LHS} = \frac{1}{1} = 1$$

$$\text{RHS} = \frac{2(1)}{1+1} = 1$$

$$\therefore \text{True for } n=1$$

Assume true for $n=k$:

$$\text{ie } \frac{1}{1} + \frac{1}{1+2} + \dots + \frac{1}{1+2+\dots+k} = \frac{2k}{k+1}$$

Prove true for $n=k+1$:

$$\text{ie } \frac{1}{1} + \frac{1}{1+2} + \dots + \frac{1}{1+2+\dots+k} + \frac{1}{1+2+\dots+k+1} = \frac{2(k+1)}{k+2}$$

$$\text{LHS} = \frac{1}{1} + \frac{1}{1+2} + \dots + \frac{1}{1+2+\dots+k} + \frac{1}{1+2+\dots+k+1}$$

$$= \frac{2k}{k+1} + \frac{1}{1+2+\dots+k+1} \quad \text{by assumption} \quad [1]$$

$$= \frac{2k}{k+1} + \frac{2}{(k+1)(k+2)} \quad \text{by part (i)}$$

$$= \frac{2k(k+2) + 2}{(k+1)(k+2)}$$

$$= \frac{2(k^2+2k+1)}{(k+1)(k+2)}$$

$$= \frac{2(k+1)^2}{(k+1)(k+2)}$$

$$= \frac{2(k+1)}{k+2}$$

$$[1]$$

$$= \text{RHS}$$

$\therefore$ true for $n=k+1$ when true for $n=k$

by M.I., true for all n are integers n .

$$[4]$$

$$(iii) \quad \sum_{n=2}^{\infty} \left(\frac{2k}{k+1} \right)$$

$$= 2$$

$$[1]$$

must indicate relevance in reasoning

(b) $P(x) = (x-4)(x+1)Q(x) + (ax+b)$

$P(4) = -5$
 $P(-1) = -5$

Solve simult: $a = -2$ (1) for the solving.

$b = 3$

$\therefore \text{remainder} = -2x + 3$

1 mark for stating the correct remainder polynomial.

Question 2

(a) $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x} = \frac{3}{2}$ [1] answer only

(b) (i) $\frac{1}{x} + \frac{1}{b} + \frac{1}{x} = \frac{a\beta + \alpha\beta + \beta\gamma}{a\beta\gamma}$ [1] correct algebra

[1] for both correct substitutions.

$= \frac{3}{2}$

(ii) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$ [1] as above

$= 0^2 - 2(-3/2)$

$= 3$

(c) let $f(x) = 2\sin x - x$

$f'(x) = 2\cos x - 1$

$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$

$= 2 - \frac{2\sin 2 - 2}{2\cos 2 - 1}$ [1]

$= 1.901$ (3d.p) [1]

No marks if double angle formulae have not been used

$\cos 4x = 1 - 2\sin^2 2x$

$\therefore \sin^2 2x = \frac{1}{2}(1 - \cos 4x)$ [1]

$\int \sin^2 2x \, dx = \frac{1}{2} \int (1 - \cos 4x) \, dx$
 $= \frac{1}{2} \left(x - \frac{1}{4} \sin 4x \right) + c$
 $= \frac{x}{2} - \frac{1}{8} \sin 4x + c$

$\int_0^{9/13} \frac{2 \, dx}{16 + 9x^2} = \frac{2}{9} \int_0^{9/13} \frac{dx}{\frac{16}{9} + x^2}$
 $= \frac{2}{9} \cdot \frac{3}{4} \left[\tan^{-1} \frac{3x}{4} \right]_0^{9/13}$
 $= \frac{1}{6} \left(\frac{\pi}{4} - 0 \right)$
 $= \frac{\pi}{24}$

[1] for $\tan^{-1} \frac{3x}{4}$
 [1] for correct coefficient
 [1] for final answer.

[3]

P.S. Most students got 1 mark since only 1 answer for $y=2$, $x=100$. However, to get the marks the solution should be as follows.

For $y=2$, $x=100$.
 $-\frac{100^2}{720} (1 + \tan^2 \alpha) + 100 \tan \alpha = 2$
 $-\frac{125}{9} (1 + \tan^2 \alpha) + 100 \tan \alpha = 2$
 $-125 (1 + \tan^2 \alpha) + 900 \tan \alpha = 18$
 $0 = 125 \tan^2 \alpha - 900 \tan \alpha + 143$
 $\tan \alpha = \frac{900 \pm \sqrt{900^2 - 4(143)(125)}}{2(125)}$
 $= \frac{900 \pm \sqrt{738500}}{250}$
 $\tan \alpha = 7.037$
 $\alpha = 81^\circ 54'$
 $y = -10t + 60 \sin \alpha = -\frac{50}{3} + 60 \sin \alpha$
 $t = \frac{2}{3 \sin \alpha} = \frac{2}{3 \sin 81^\circ 54'}$

✓ both answers 81°54' and 90°13'

Question 3

1) (i) Alternate segment theorem [1]

(ii) $\angle CDB = \angle TDB$ (alt. seg. thm)

$= \angle TDP$ (vert. opp. $\angle$ s) [1]

$= \angle BPT$ (alt. seg. thm) [1]

$\therefore AB \parallel CD$ (alternate angles equal) [1] [3]

(b) (i) $\cos x - \sqrt{3} \sin x = R \cos(x + \alpha)$

$$= R(\cos x \cos \alpha - \sin x \sin \alpha)$$

$$= (R \cos \alpha) \cos x - (R \sin \alpha) \sin x$$

$$\therefore R \cos \alpha = 1 \quad \text{--- (1)}$$

$$R \sin \alpha = \sqrt{3} \quad \text{--- (2)}$$

$$\text{(1)}^2 + \text{(2)}^2: R^2 = 4 \Rightarrow R = 2$$

$$\text{(2)} \div \text{(1)}: \tan \alpha = \sqrt{3} \Rightarrow \alpha = \frac{\pi}{3} \quad (\text{1st quad})$$

$$\therefore \cos x - \sqrt{3} \sin x = 2 \cos\left(x + \frac{\pi}{3}\right) \quad [2]$$

(ii) $\cos x - \sqrt{3} \sin x = \sqrt{3}$

$$2 \cos\left(x + \frac{\pi}{3}\right) = \sqrt{3}$$

$$\cos\left(x + \frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

$$x + \frac{\pi}{3} = \frac{\pi}{6}, \frac{5\pi}{6} \quad [1]$$

$$x = -\frac{\pi}{6}, \frac{2\pi}{3}$$

$$x = \frac{3\pi}{2}, \frac{11\pi}{6} \quad [1]$$

$$[2]$$

Question 6

(i) (i) $\ddot{x} = 0$

$$x = c \quad \ddot{y} = -10 \quad \ddot{y} = -10 + c$$

$$= 60 \cos \alpha \quad [1] \quad = -10 + 60 \sin \alpha \quad [1]$$

$$x = 60 \cos \alpha + c \quad y = -5t^2 + 60 \sin \alpha + c$$

$$= 60 \cos \alpha \quad [1] \quad = -5t^2 + 60 \sin \alpha \quad [1]$$

$$[4]$$

(ii) $x = 60 \cos \alpha \Rightarrow t = \frac{x}{60 \cos \alpha}$

$$y = -5t^2 + 60 \sin \alpha$$

$$= -\frac{5x^2}{3600 \cos^2 \alpha} + 60 \cdot \frac{x}{60 \cos \alpha} \cdot \sin \alpha \quad [1]$$

$$= -\frac{1}{720} x^2 \sec^2 \alpha + x \tan \alpha$$

$$= -\frac{1}{720} x^2 (1 + \tan^2 \alpha) + x \tan \alpha \quad [1] \quad [2]$$

(iii) height of Harold above ~~ground~~

$$\text{point of projection} = \frac{1}{10} \times 100 + 2$$

$$= 12 \text{ metres}$$

! for correct
y=12 into
Equation.

$$x = 100, y = 12 \Rightarrow 12 = -\frac{1}{720} \times 100^2 (1 + \tan^2 \alpha) + 100 \tan \alpha \quad [1]$$

$$12 = -\frac{125}{9} (1 + \tan^2 \alpha) + 100 \tan \alpha$$

$$108 = -125 - 125 \tan^2 \alpha + 900 \tan \alpha$$

$$125 \tan^2 \alpha - 900 \tan \alpha + 233 = 0$$

$$\tan \alpha = \frac{900 \pm \sqrt{900^2 - 4 \times 125 \times 233}}{250}$$

$$= 6.93107, 0.26894$$

$$\alpha = 15^\circ 41', 81^\circ 47' \quad [1]$$

$$0.268, 1.417$$

! for clarity
solving the Q. E
! for better
answers

By
tasting: ~~see~~ Check: $t = \frac{100}{60 \cos \alpha} = \frac{5}{3 \cos \alpha}$

$$y = -\frac{50}{3 \cos^2 \alpha} + 60 \sin \alpha$$

$$\alpha = 15^\circ 41', y = -1.66 \text{ m/s}$$

$$\alpha = 81^\circ 47', y = -57.23 \text{ m/s}$$

$$[3]$$

b) (i) $T = 20 + Ae^{-kt}$

$$\frac{dT}{dt} = -kAe^{-kt}$$

$$= -k(5-20) \quad \underline{\underline{[1]}}$$

(ii) $T = 20 + 75e^{-kt}$

$$70 = 20 + 75e^{-10k}$$

$$e^{-10k} = \frac{2}{3}$$

$$k = -\frac{1}{10} \ln \frac{2}{3} = 0.04058 \quad \underline{\underline{[1]}}$$

$$t = 15, T = 20 + 75e^{-15k}$$

$$= \cancel{60.88} 61^\circ\text{C} \quad \underline{\underline{[2]}}$$

(c) (i) $2\sin 2A \cos 2A = \sin 4A \quad \underline{\underline{[1]}}$

(iii) $\text{LHS} = \frac{\sin 3A}{\sin A} + \frac{\cos 3A}{\cos A}$

$$= \frac{\sin 3A \cos A + \cos 3A \sin A}{\sin A \cos A} \quad \underline{\underline{[1]}}$$

$$= \frac{\sin(3A+A)}{\sin A \cos A} \quad \underline{\underline{[1]}}$$

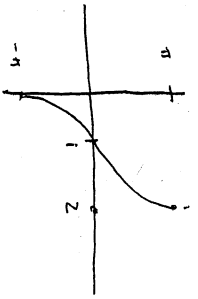
$$= \frac{2\sin 2A \cos 2A}{\frac{1}{2} \sin 4A} \quad \text{from part (i)}$$

$$= 4 \cos 2A \quad \underline{\underline{[1]}}$$

$$= \text{RHS} \quad \underline{\underline{[3]}}$$

Question 4

1) $\frac{d}{dx} \left(\frac{-x^2 + 7x - 12}{x^2} \right)$ [2]



wrong shape = 0
deduct one mark for each of the following incorrect: domain, range, orientation.

2) (i) $\ddot{x} = \frac{d}{dx} \left(\frac{-x^2 + 7x - 12}{x^2} \right)$

$= -x + \frac{7}{x}$ [1]

(ii) $\ddot{x} = -1 \left(x - \frac{7}{x} \right)$ [1]

$\therefore$ centre is $x = \frac{7}{2}$ [1]

(iii) $\dot{x}^2 = 1 \Rightarrow \dot{x} = 1$ [1]

$\therefore$ period = $\frac{2\pi}{1} = 2\pi$ sec. [1]

(c) $\frac{2}{x-1} < x$

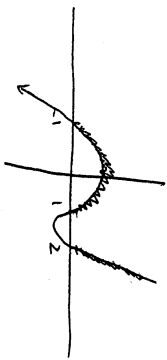
$2(x-1) < x(x-1)^2$ [1]

$x(x-1)^2 - 2(x-1) > 0$

$(x-1)[x(x-1)-2] > 0$

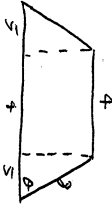
$(x-1)(x^2-x-2) > 0$

$(x-1)(x-2)(x+1) > 0$



$-1 < x < 1$ or $x > 2$ [2]

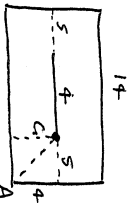
d) (i)



$\cos \theta = \frac{5}{8}$

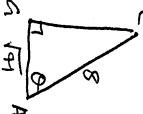
$\theta = 51.94^\circ$ [2] 1 for diagram showing the S.

(ii) Let A be pentagon base vertically below C.



$CA^2 = 5^2 + 9^2$

$CA = \sqrt{141}$ [1]



$\cos \theta = \frac{\sqrt{141}}{8}$

$\theta = 36.50^\circ$ [1]

[2]

Question 5

a) (i) $x = 4p$ $y = 2p^2$

$\frac{dx}{dp} = 4$ $\frac{dy}{dp} = 4p$

$\frac{dy}{dx} = \frac{4p}{4} = p$ [1]

OR

$x = 4p \Rightarrow m_T = p$

$y - 2p^2 = p(x - 4p)$

$y - 2p^2 = px - 4p^2$

$y = px - 2p^2$ [1]

(ii) at Q, $p = -1$

$\therefore$ tangent: $y = -x - 2$ [1]

(iii) $y = px - 2p^2$

$y = -x - 2$

$-px - 2p^2 = -x - 2$ [1]

$px + x = 2p^2 - 2$

$x(p+1) = 2(p-1)(p+1)$

$x = 2(p-1)$

$= 2p - 2$ [1]

$\therefore T(2p-2, -2p)$

(iv) $P(4p, 2p^2)$, $T(2p-2, -2p)$

$M \left[\frac{4p+2p-2}{2}, \frac{2p^2-2p}{2} \right] \Rightarrow M(3p-1, p^2-p)$ [1]

i.e. $x = 3p-1 \Rightarrow p = \frac{x+1}{3}$ [1]

$y = p^2 - p$

$= \left(\frac{x+1}{3} \right)^2 - \frac{x+1}{3}$

$= \frac{(x+1)^2 - 3(x+1)}{9}$

$y = (x+1) \left[\frac{x+1-3}{3} \right]$

$y = (x+1)(x-2)$ [1]

[3]

If they do it the long way, give them the mark.

[2]